

INVESTIGATION REPORT

Issue 1015: Operand-select combination in the ID stage

openhwgroup/cv32e40p

Document	RTL-TRIAGE-CV32E40P-1015
Disposition	Dead code, proven
Record date	2026-09-18
Classification	Internal working document. Not a verification sign-off.

1 Status

Issue	openhwgroup/cv32e40p#1015
Issue type	Coverage hole referred for disposition
Affects	cv32e40p
Configuration	RTL master 6033d2b, COREV_PULP=1 FPU=0
Disposition	Dead code, proven
Severity	Not assessed
Agreement	Same disposition as the independent human record.
Evidence	5 cited, 3 present, 2 not retained

Disposition meaning. The code cannot be reached in any state permitted by the stated model and assumptions, established by an unbounded proof.

Severity is not recorded for this run. The harness does not infer a severity from the disposition, so the field is reported as not assessed rather than filled in.

2 Description

The uncovered combination of the if at cv32e40p_id_stage.sv:872 (case branch OP_C_REGC_OR_FWD of the APU read-register logic) is unreachable. The if has five atoms: `alu_op_a_mux_sel!=OP_A_REGC_OR_FWD`, `ctrl_transfer_target_mux_sel!=JT_JALR`, `alu_op_b_mux_sel==OP_B_IMM`, `alu_bmask_b_mux_sel==BMASK_B_REG`, `alu_op_b_mux_sel==OP_B_BMASK`. The only focused-expression row with no instruction behind it is the one where `alu_op_b_mux_sel==OP_B_IMM` is false while `alu_bmask_b_mux_sel==BMASK_B_REG` is true and every other atom is non-masking, i.e. bmask taken from a register while operand B is neither an immediate nor a bitmask and operand A is not rC. The decoder sets `BMASK_B_REG` at exactly four places (cv.insert with `OP_B_BMASK`, cv.bclrr/cv.bset with `OP_B_IMM`, cv.addNr/subNr family with `OP_A_REGC_OR_FWD`), so that row cannot occur. An unbounded proof (abc pdr, top cv32e40p_core, COREV_PULP=1) of the exact row plus the decoder invariant `BMASK_B_REG => (OP_B_IMM | OP_B_BMASK | OP_A_REGC_OR_FWD)` passes with FPU=0 and again with FPU=1 where `gen_apu` and line 872 are actually elaborated. A bounded cover run reaches the enclosing branch at step 2 and each of the four neighbouring legal rows (cv.bclrr/bset, cv.addNr, cv.insert, JALR) at step 6, so the proof is not vacuous and every other row of the expression is reachable. Environment assumptions: legal OBI handshakes, `scan_cg_en_i=0`, `fetch_enable_i=1`; instruction memory contents unconstrained.

3 Conditions

1. legal OBI handshakes
2. `scan_cg_en_i=0`
3. `fetch_enable_i=1`
4. instruction memory contents unconstrained

Extracted verbatim from the environment assumptions stated in the verdict record for this run.

4 Implications

Not recorded as a separate field for this run. What the disposition means for a user of the core follows from the Disposition meaning in Status and the bounds in Limitations and scope.

5 Action

Waiver text, to be applied to the coverage exclusion

cv32e40p_id_stage.sv:872 (gen_apu, case OP_C_REGC_OR_FWD): the expression row with alu_bmask_b_mux_sel==BMASK_B_REG and alu_op_b_mux_sel not OP_B_IMM (and not OP_B_BMASK, alu_op_a_mux_sel not OP_A_REGC_OR_FWD, not JALR) is formally proven unreachable (abc pdr, unbounded, top cv32e40p_core, COREV_PULP=1, checked with FPU=0 and FPU=1). The decoder drives BMASK_B_REG only for cv.insert (OP_B_BMASK), cv.bclrr/cv.bsetr (OP_B_IMM) and the cv.addNr/subNr family (OP_A_REGC_OR_FWD), so the atom alu_op_b_mux_sel==OP_B_IMM can never be the sole controlling term. Evidence: runs/v1-cv32e40p-1015/formal/{p_assert,p_assert_fpu1}/sby.log (PASS) and formal/c_state/sby.log (5/5 neighbouring rows covered). Exclude that FEC row only; all other rows of line 872 are reachable.

Checker used

checker_1015_assert.sv

6 Evidence

Ref	Type	Path, relative to the run directory	File	Result recorded
E1	proof	formal/p_assert/sby.log	present	abc pdr PASS (FPU=0 COREV_PULP=1): exact FEC row of line 872 never true in branch OP_C_REGC_OR_FWD; decoder invariant BMASK_B_REG => OP_B_IMM OP_B_BMASK OP_A_REGC_OR_FWD holds
E2	proof	formal/p_assert_fpu1/sby.log	present	abc pdr PASS (FPU=1): same checker with gen_apu (line 872) elaborated; yosys log lists cv32e40p_apu_disp
E3	proof	formal/c_state/sby.log	present	smtbmc/yices cover PASS (depth 40): covers_reached=5 - branch alone at step 2; bclrr/bsetr, addNr, insert and JALR rows at step 6 -> asserts not vacuous
E4	counterexample	formal/c_state/job/engine_0/trace1.vcd	not kept	cover witness: instr 0x3a00302b (cv.bclrr) gives OP_C_REGC_OR_FWD + BMASK_B_REG + OP_B_IMM; rendered in cover_bclrr_row.png
E5	counterexample	formal/c_state/job/engine_0/trace2.vcd	not kept	cover witness: cv.addNr family gives OP_C_REGC_OR_FWD + BMASK_B_REG + OP_A_REGC_OR_FWD; rendered in cover_addNr_row.png

Paths are relative to runs/v1-cv32e40p-1015. The File column is a check made when this document was generated. It reports whether the cited file exists, not whether its contents were re-validated. 'not kept' means a repository rule deliberately excludes that artefact, usually a waveform dump; rerun the check to regenerate it. 'MISSING' means the file is absent and was expected to be there.

What each class of artefact establishes

- proof. Proof or engine run. Scope depends on the mode: prove is unbounded under the stated model, bmc is bounded to its depth, cover establishes reachability only.
- counterexample. A concrete input sequence produced by the engine. Shows the stated condition is reachable; it is not itself a proof.

Artefact(s) E4, E5 are not retained under repository policy. The result recorded against them stands on the engine log cited alongside; the artefact itself must be regenerated by rerunning the check before it can be inspected.

- 7

Limitations and scope

 - The proof establishes unreachability under the model and the environment assumptions named in Status and Conditions. It does not establish that the surrounding logic is correct.
 - This is an internal investigation record. It requires engineering review before it is used as a disposition. No report generator can supply missing evidence.
 - Results apply to the configuration and core revision named in Status. A different configuration needs a separate run.
 - Public demonstration runs on open cores are not client engagements and not production verification results.

8

Provenance and measurement

Run directory	runs/v1-cv32e40p-1015
Title source	Title and configuration read back from reports/issues/cv32e40p_issue_1015.pdf, the previously published report for this issue.
Inputs shown	title, body(links stripped), labels
Human record	Dolphin Design 2024: waived (ID stage line 872)
Assessed confidence	0.92. This is the investigator's own estimate. It is not a measurement and not a reliability figure.
Iterations	1

Generated by reporting/scripts/issue_document.py from runs/v1-cv32e40p-1015/verdict.json. The generator typesets recorded data and checks that cited files exist. It does not run, rerun or validate any check, and it cannot supply missing evidence.